

Code:

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <unistd.h>
5 #include <fcntl.h>
6 #include <sys/stat.h>
7 #include <errno.h>
8
9 #define FIFO1 "client_to_server"
10 #define FIFO2 "server_to_client"
11
12 int main()
13 {
14     int fd1, fd2;
15     char message[100];
16     char response[200];
17     ssize_t bytes_read;
18
19     // Create FIFOs
20     if (mkfifo(FIFO1, 0666) == -1 && errno != EEXIST)
21     {
22         perror("mkfifo client_to_server");
23         return 1;
24     }
25
26     if (mkfifo(FIFO2, 0666) == -1 && errno != EEXIST)
27     {
28         perror("mkfifo server_to_client");
29         return 1;
30     }
31
32     printf("Server started...\n");
33     printf("Waiting for client message...\n");
34
35     // Open FIFO for reading
36     fd1 = open(FIFO1, O_RDONLY);
37
38     if (fd1 == -1)
39     {
40         perror("open FIFO1");
41         return 1;
42     }
43
44     // Read message
45     bytes_read = read(fd1, message, sizeof(message) - 1);
46
47     if (bytes_read == -1)
48     {
49         perror("read");
50         close(fd1);
51         return 1;
52     }
53
54     // Add null character
55     message[bytes_read] = '\0';
```

```

56
57     printf("Client: %s\n", message);
58
59     close(fd1);
60
61     // Process message
62     snprintf(response, sizeof(response),
63              "Server received and processed: %s", message);
64
65     // Open FIFO for writing
66     fd2 = open(FIFO2, O_WRONLY);
67
68     if (fd2 == -1)
69     {
70         perror("open FIFO2");
71         return 1;
72     }
73
74     // Send response
75     write(fd2, response, strlen(response) + 1);
76
77     printf("Response sent to client.\n");
78
79     close(fd2);
80
81     // Remove FIFOs
82     unlink(FIFO1);
83     unlink(FIFO2);
84
85     return 0;
86 }
87

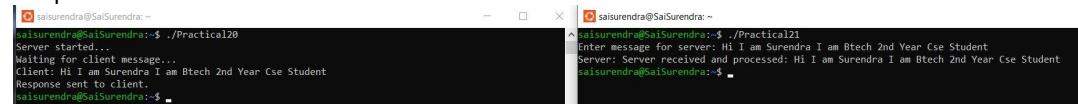
```

```

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2  #include <stdlib.h>
3  #include <string.h>
4  #include <unistd.h>
5  #include <fcntl.h>
6
7  #define FIFO1 "client_to_server"
8  #define FIFO2 "server_to_client"
9
10 int main()
11 {
12     int fd1, fd2;
13     char message[100];
14     char response[100];
15
16     printf("Enter message for server: ");
17     fgets(message, sizeof(message), stdin);
18
19     // Remove newline
20     message[strcspn(message, "\n")] = '\0';
21
22     // Open FIFO for writing
23     fd1 = open(FIFO1, O_WRONLY);
24
25     // Send message to server
26     write(fd1, message, strlen(message) + 1);
27
28     close(fd1);
29
30     // Open FIFO for reading
31     fd2 = open(FIFO2, O_RDONLY);
32
33     // Read server response
34     read(fd2, response, sizeof(response));
35
36     printf("Server: %s\n", response);
37
38     close(fd2);
39
40     return 0;
41 }
42

```

Output:



The image shows two terminal windows side-by-side. The left window, titled 'saisurendra@SaiSurendra: ~', shows the execution of './Practical20'. It displays the server starting, waiting for a client message, receiving the message 'Hi I am Surendra I am Btech 2nd Year Cse Student', and sending a response. The right window, also titled 'saisurendra@SaiSurendra: ~', shows the execution of './Practical21'. It displays the server entering a message, receiving the same client message, and processing it.

```
saisurendra@SaiSurendra:~$ ./Practical20
Server started...
Waiting for client message...
Client: Hi I am Surendra I am Btech 2nd Year Cse Student
Response sent to client.
saisurendra@SaiSurendra:~$
```

```
saisurendra@SaiSurendra:~$ ./Practical21
Enter message for server: Hi I am Surendra I am Btech 2nd Year Cse Student
Server: Server received and processed: Hi I am Surendra I am Btech 2nd Year Cse Student
saisurendra@SaiSurendra:~$
```